

Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests demonstrate values entirely within reference ranges, indicating preserved hepatic synthetic and metabolic function. **ML Result: Normal** – the predictive model assesses the profile as within normal limits.

Detailed Test Explanations

Serum Bilirubin (Total): Measures overall bilirubin levels, a byproduct of red blood cell breakdown processed by the liver; elevated levels suggest hepatocellular dysfunction, biliary obstruction, or hemolysis. Patient result: 0.9 mg/dL (reference: 0.2-1.2 mg/dL) – normal.

Serum Bilirubin (Direct): Quantifies conjugated bilirubin excreted via bile; elevations indicate biliary obstruction or intrahepatic cholestasis. Patient result: 0.2 mg/dL (reference: 0-0.3 mg/dL) – normal.

Serum Bilirubin (Indirect): Reflects unconjugated bilirubin; increases may signal hemolysis or impaired hepatic uptake/conjugation. Patient result: 0.70 mg/dL (reference: 0.2-1.0 mg/dL) – normal.

SGPT (ALT): Alanine aminotransferase, a sensitive enzyme marker of hepatocellular injury; elevations occur with acute hepatitis or toxic insults. Patient result: 36 U/L (reference: 13-40 U/L) – normal.

SGOT (AST): Aspartate aminotransferase, present in liver, heart, and muscle; rises with hepatic necrosis or extrahepatic sources. Patient result: 32 U/L (reference: 0-37 U/L) – normal.

Serum Alkaline Phosphatase: Enzyme elevated in cholestasis, bone disease, or biliary obstruction; result unavailable.

Serum Protein: Total protein reflecting hepatic synthetic capacity and nutritional status; hypoalbuminemia or hypoproteinemia signals chronic liver disease. Patient result: 7.2 g/dL (reference: 6.4-8.3 g/dL) – normal.

Serum Albumin: Primary protein synthesized by hepatocytes, indicator of nutritional and liver synthetic function; low levels denote chronic liver disease or malnutrition. Patient result: 4.7 g/dL (reference: 3.5-5.2 g/dL) – normal.

Globulin: Immunoglobulin fraction; alterations suggest inflammation, infection, or dysproteinemia. Patient result: 2.50 g/dL (reference: 1.8-3.6 g/dL) – normal.

A/G Ratio: Albumin-to-globulin ratio; imbalance may indicate chronic liver disease or inflammatory states. Patient result: 1.88 (reference: 1.1-2.4) – normal.

Overall Interpretation

All performed liver function parameters fall within normal reference intervals, consistent with intact hepatocellular integrity, bilirubin metabolism, and protein synthesis. The normal ML Result reinforces absence of predictive abnormal patterns; clinical correlation advised per laboratory remarks.

Recommendations

Maintain balanced diet rich in fruits, vegetables, and lean proteins; limit alcohol intake; engage in regular aerobic exercise (150 minutes weekly); monitor for symptoms like jaundice or fatigue. Routine annual hepatic screening recommended; consult physician for personalized follow-up.

Disclaimer

This report is for informational purposes only and does not constitute medical advice. Always consult a qualified healthcare provider for diagnosis and treatment.